

Linear A at 2026: what is known, what has been tested, and what the corpus cannot decide

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Summary

A century after Evans, Linear A is undeciphered and the field has no shared account of why. Reviews of the script list proposals and repeat findings; none states which claims have been tested against a null, which fail, and which cannot be decided with the corpus that exists. This paper is that account. It reviews the state of the question across phonology, morphology, lexicon, syntax, the writing system and the archive; it reports the measurement of twelve claims the field currently holds, of which three are confirmed, three are narrowed, one is refuted and one turns out to hold for a single sign; and it names the walls the corpus imposes, each with a figure. Two of those walls are new: **76% of the corpus dates to a single destruction horizon**, so there is no temporal series to analyse; and **102 identified scribal hands share a mean vocabulary overlap of 0.026**, so the corpus is not 552 units of one language but many small and nearly disjoint vocabularies. Together with an information count that puts the deficit at 345 bits, and a decipherment model validated on Linear B that requires half the sign inventory anchored before it recovers anything, they explain a century of failure without appeal to method.

Keywords: Linear A, Minoan, undeciphered scripts, decipherment, null models, state of the art.

1. Why a review with verdicts

The literature on Linear A is large and its findings circulate by citation. KU-RO means total; SA-RA₂ and TE are mutually exclusive and therefore denote opposite transactions; the entries of HT 86 and HT 95 form a connected group; Minoan barely uses the o-series. Some of these are right, some are right in a narrower sense than stated, and at least one is an inference from a fact that carries no information.

Nothing in the field distinguishes them, because nothing in the field measures them. A review that repeats the claims reproduces that problem. What follows reports, for each, what a null model says.

The same applies to the negative side. “Linear A is not inflected”, “there is no lexical continuity with Eteocretan”, “the Cretan glosses preserve Minoan vocabulary” are statements that require a power analysis before they mean anything, and in two of those three cases the power is zero.

2. The corpus, and two walls

1,720 documents, some 7,150 signs, 552 intact syllabic units, 110 signs. The largest archive is Haghia Triada with 199 documents; Khania has 121, Zakros 44.

Two unit counts recur in this series and are different quantities: **552** intact syllabic units in the whole corpus, which is what the information count uses, and **443** in the accounting documents

alone, of which 389 occur on tablets only. Figures quoted below follow whichever is proper to the measurement.

The first wall is chronological. 1,308 of the 1,720 documents, **76%**, belong to the LM IB destruction horizon. The corpus is a photograph of one year, not a series. No linguistic change, no palaeographic evolution and no internal dating of variants can be measured, because there is nothing to measure them against.

The second is scribal. 102 hands are identified. Of the fourteen with ten or more intact units, the mean Jaccard overlap of their vocabularies is **0.026**, and the highest pair, two scribes of Zakros, share five units out of eighteen and thirteen. Each scribe writes about different things. The corpus is not 552 units of a language but a set of small, nearly disjoint vocabularies, which is why tests of recurrence have so little power.

Those two walls, and not the absence of ideas, are why the statistics of this corpus behave as they do.

3. Phonology: four facts, each with its control

	Minoan	control	power
o-series	4.7% of vowels, 0.9% without RO and KO; eight o-signs with zero occurrences	Mycenaean 26.3%	—
u-series	17.4%	Mycenaean 7.7%	—
vowel harmony	absent: 27.1% against 27.8%, $p =$ 0.74	Linear B has it: 29.3% against 24.0%, $p < 0.0001$	100% , 30 of 30 subsamples
dependency between adjacent syllables	consonant excess 0.081, $p < 0.005$	Linear B at the same size: 0.226	high
reduplication	4.4% of units	Mycenaean 2.3%, $p =$ 0.0038	—
-u / -o doublets with Linear B	16 of 90	null 8.0, $p < 0.002$	—

A vowel system poor in o and rich in a and u, reduplication at twice the Greek rate, no harmony, and weak dependency between adjacent syllables. None of these requires a candidate language. Any proposed reading must satisfy all four, and **no candidate in the literature satisfies the absence of o together with the abundance of e** — Luwian lacks the phoneme e, and the Semitic languages have o.

4. Morphology: undeterminable, and for two independent reasons

Kober established that Linear B was inflected by finding ending alternations recurring across stems. On Linear A the same test gives **4 bridges against 6.6 expected, $p = 0.86$** .

That negative says nothing, and the power analysis shows why. Linear B subsampled to the 501 units Linear A has gives a bridge ratio of 0.56 against Linear A's 0.61: indistinguishable. Four inflection statistics, over 25 subsamples each, detect Greek inflection at that corpus size in **zero cases**.

And a second reason is structural, not statistical. If Eteocretan descends from Minoan and admits final -t and -d, a CV syllabary cannot write those consonants, exactly as Linear B omits the final consonants of Greek. The endings that are measured are the last written syllable, not the end of the word. **Part of Minoan morphology is invisible by construction of the script.**

5. Lexicon and language: eight tests, eight closed

test	result
thirteen candidate languages, anchored value search	none reaches $z = 3$; highest Etruscan +1.75; Mycenaean control -0.90
double-constraint screening	zero matches in Akkadian, Ugaritic and Sumerian over 44,855 glossed entries
search with the functional constraint as sole objective	1 of 20, and spurious
the acronym hypothesis	2 of 11, null 1.64, $p = 0.52$
Linear B toponyms	3 of 17 appear ($p = 0.0005$); the other fourteen are absent from the corpus
non-Greek vocabulary of Linear B	2 of 19, and both are inherited signs , not loans
Cretan glosses of Hesychius	37 non-Greek; none testable against an ideogram
Eteocretan	0 of 71 under the strict method

And the shape of the failure is symmetrical. Under the permissive phonology a CV syllabary imposes, 89.1% of Linear A units find a Semitic comparandum, Greek finds one 79.7% of the time, and **fictitious grids reach 97.1%**. Optimising for lexical fit finds grids 2.4 times better than the standard one. By lexicon there is too much evidence; by typology there is too little, and three structural features give three different answers.

6. Syntax

Prior art: Davis (*Kadmos* 52, 2014) established verb-subject-object order from the libation formula; Younger identified transaction words by position; Schoep classified documents by content.

Measured here, on the accounting document:

Each unit carries its own figure. UNIT → NUM runs at 1.41 over the null in Linear A and **0.70 in Linear B**, where units chain before reaching the number. UNIT → UNIT is 0.63 against 1.07. The comparison uses 2,884 transitions against 21,778.

The label opens and the total closes: mean positions 0.300 and 0.737 against a null of 0.506, both $p < 0.0005$. **And the deficit does not close:** KI-RO sits at 0.498, the exact centre, $p = 0.89$. It is a line within the account, not its balance.

Davis's fixed order is confirmed on the 35 stone-vessel inscriptions with three or more words: A-TA-I-*301-WA-JA opens in 7 of 7 (mean position 0.00), SI-RU-TE closes (0.96). The verb-subject-object identification is not confirmed here, since assigning those categories requires knowing what the words mean.

7. The writing system

Ligatures bind a syllabogram to a commodity, and the syllabogram is specific to it: 16 of 26 go with one commodity only, against 25% by chance, $p < 0.0005$. **But the ligature does not constrain the phonetic value:** tested on Linear B, acrophony of the commodity gives 2 of 10 ($p = 0.22$) and abbreviation of the adjacent qualifier 4 of 11 ($p = 0.41$).

The signs Greek did not inherit open their unit more than the rest: 46.8% against 32.3%, with a frequency-matched null of 29.3%, $p = 0.002$, and they appear on roundels 19% of the time against 5%. What the Mycenaeans discarded was the vocabulary of heading and sealing, not the vocabulary of the count.

Two signs carry both sound and sense, which is what an anchor requires: **NI**, the fig ideogram, taken from Linear A and read by Neumann (1962) as the initial syllable of a Minoan word preserved in the Greek gloss *nikuleon*; and the goat ideogram with value *mi* against the Cretan gloss *miglas*. Three further ligatures — TELA+KU, OLE+RA, CYP+PA — occur in both scripts with no Greek word to explain them, and are the open question this work leaves to specialists.

8. Twelve claims of the field, measured

claim	source	verdict
the entry groups of HT 86 and 95	Packard 1974; Uchitel & Finkelberg 1995	confirmed , 35 pairs against 8.2
breakdown of the o-series	Finkelberg 2001 n.10	12 of 13 exact , 31 years on
-u / -o doublets	Lejeune 1956	confirmed against a null for the first time

claim	source	verdict
reduplication as a Minoan trait	Lejeune 1956	confirmed , $p = 0.0038$
<i>ku-ni-su</i> heads HT 10a	Uchitel & Finkelberg 1995	confirmed
<i>ki-ki-ra-ja</i> heads HT 85b	idem	confirmed
SA-RA ₂ and TE are mutually exclusive	Schoep	the fact yes, the inference no : $p = 0.50$
PO-TO-KU-RO sums the KU-RO	Schoep	holds in one of two
KI-RO is less than KU-RO	Schoep	holds, but only two pairs exist
<i>a-du</i> as an Anatolian imperative	Finkelberg 2001	refuted : the paradigm opens the document in 12 of 17, $p = 0.012$
transaction signs stand between dividers	Younger	true for TE alone , 13 of 58, $p = 1.2 \cdot 10^{-15}$; KA gives 0 of 169
word order of the libation formula	Davis 2014	fixed order confirmed ; the V-S-O labels not tested here

9. What the corpus cannot decide, and its price

Two independent counts, agreeing to within 2%:

	direct count	decipherability function
cost of specifying a decipherment	587 bits	587
what the corpus supplies	236 (lexical fit)	242 (measured match rate 0.738)
deficit	351 bits	345 bits

In the currencies a field can act on: **56 signs to anchor from outside**, against the 16 there are, or some **788 further intact units** of text, against the 552 there are.

Confirmed by an independent route. The model of Luo and colleagues (2021), extended here to syllabaries and validated on Linear B, **takes off between 7 and 10 anchors of twenty signs**. Linear A sits at 2.9 on that scale.

10. The walls, and what would remove each

wall	figure	what removes it
no temporal series	76% of the corpus from one destruction	excavation of pre-LM IB deposits
disjoint scribal vocabularies	mean Jaccard 0.026 across 102 hands	more text per hand, which means more text
power floor	four inflection statistics detect Greek at zero% when given 501 units	roughly doubling the intact corpus
invisible final consonants	structural property of a CV syllabary	nothing internal; a bilingual or a related script that writes them
the lexical criterion does not discriminate	fictitious grids at 97.1%	anchors from outside, not a better search
Cretan Hieroglyphic does not help	its 331 texts are published as photographs; the sign index gives no order	transcription of the plates by a specialist, weeks of trained work
56 anchors missing	345 bits	a bilingual, or new documents naming known places or persons

None of the seven is removed by computation. Five require excavation or specialist transcription; two require a find.

11. What this work leaves

A functional inventory of 151 readings — 67 fixed unit by unit, the rest by the class function of the opening position — each with evidence for, evidence against and a stated refutation condition.

Thirty closed avenues and twenty-one withdrawn claims, each with its measurement and its date. Six of the withdrawals were positives of this work that a control destroyed, among them a z of +6.88 for Akkadian that turned out to be editorial marks in the skeletons, and a sign that opened its unit in 15 of 15 occurrences which were all roundels of a single Khania series.

Thirty-two falsifiable predictions about the Minoan lexicon: for each ligature whose syllabogram carries an inherited value, the Minoan word for that commodity contains that syllable. And two more from the Cretan glosses: if the value of SUS is ever fixed it must be /ma/, and if VAS is fixed, /a/.

And a protocol of ten minimum requirements, calibrated on four corpora, whose fourth requirement no decipherment proposal in the literature meets: state how many tests were run before the one that is published.

12. What this does not claim

Not one phonetic value is fixed here that was not already known. The inventory establishes what units **do**, never what they **mean**, and no quantity of the first produces the second: Etruscan is fully read and only partly understood.

The conclusion is not that Linear A cannot be deciphered. It is that **the corpus does not contain enough information to distinguish the right assignment from the many wrong ones**, which is a weaker and more precise statement, and one that names its own price.

Statement of artificial intelligence assistance

The measurements and the drafting of this paper were carried out with the assistance of a language model (Claude, Anthropic), under the design, verification and responsibility of the author.